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Centellaasiatica L. - A valuable and remedial herb of santhal paragana division used in treatment as antioxidant, antimicrobial, anti-inflammatory, cytotoxic, neuroprotective

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Abstract : The present research work has been conducted to investigate one of the aquatic imperative herb viz. *centella asiatica* Linn. (*Hydricityl asiatica* L.), commonly known as Thankuni or Falguni in Santhal Paragana division of Jharkhand. The purpose of research is to explore the ethno botanical studies, medical utilization and nutritional value of this plant. Besides these, investigation was also done to survey common use of this flora as "Thankuni Saag" by the local tribal people of research area. Presence of CENTELIN at moderate temperature also plays important role in curing a number of diseases in the shape of traditional medicine by the poor mass. Information's were collected from the local tribal people and desi herbal medicine sellers in the village market pertaining to use of *Centella asiatica* Linn. in daily life. It is a perennial herb that is most commonly used by the local people due to its curative properties for various ailments and very rich nutritional value. The present work compiles an account of documentation and brief floristic description of *Centella asiatica* L. and it's traditional medicinal values and monetary benefit to the native people.

Keywords : *Centella asiatica*, Thankuni , Ethno botanical , Nutritional, Medicine.

INTRODUCTION

Centella asiatica syn. *Hydrocotyl asiatica* Linn. is a perennial angiospermic herb belonging to family Apiaceae , previously known as Umbelliferae has faintly aromatic and medicinal value since ancient period. According to Charaka Chikitsa, the juice of this herb promotes longevity and cures all kinds of diseases. It is widely distributed throughout tropical and subtropical region of the world particularly during winter seasons at damp and marshy places. The use of this significant flora as food and medicine has been increased over the years basically due to it's antioxidant, antimicrobial, anti-inflammatory, cytotoxic, neuroprotective and many more mode of action. It is one of the most popular herb in Indian medicinal system having ingredients of several compound formulations, most commonly used in the treatment of disorders related to central nervous system, skin, gastrointestinal and menstruation. It has been used since rigvedic period for improving memory, vigor and vitality. It is also being used frequently by the local tribal communities of Santhal Paragana division as vegetable (saag) due to its rich nutritive value. It is an excellent food to increase immunity and solve certain problems of ageing. This herb is commonly known as "Thankuni" or "Falguni patta" by the local people.

This remedial herb contains an alkaloid, "Centelin" at moderate temperature which plays very significant role in curing a number of diseases. The leaves of these plants are roasted and given in infusion to children in bowel complaints and fever. They are also applied to bruised parts as anti-inflammatory. This flora seems to be boon to mankind playing excellent role in leprosy and other skin diseases. It is a good tonic, immunity booster and blood purifier. Sometimes it may produce narcotic effect and produces headache and insensibility. Leaves juice and root powder is used commonly in stomach trouble of babies. Leaves juice is also employed in amoebic dysentery, diarrhea, injuries and syphilis skin diseases. Due to it's multifarious uses, this plant has great demand in Indian as

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well as in International market.

This plant is socially desirable due to it's high pharmacological value and it has been treated as an elixir of human life. This herb can also be used in earning a lot by creating favorable market prospects. The socio – economic status of tribal's may be upgraded by commercial cultivation of this plant. This will be truly a milestone in eradicating the unemployment problem among the new generations and checking them to enter in the field extremist activities. Santhal pargana division is dominated by santhal tribes and Agriculture is the main occupation for them. But local tribal communities are unable to fulfill their needs and they face the problem of seasonal unemployment. Commercial cultivation and marketing of this herb may create an opportunity for them to improve their socio–economic status. It is economically affordable and sustainable.

In spite of having so many important role, this plant species has been markedly depleted because of its unrestricted exploitation and limited cultivation. International Union of Conservation of Nature and Natural Resources (IUCN) has listed *centella asiatica* L. in the group of Threatened and Endangered species (Singh et al 2010). But National Medicinal Plant Board (NMPB) of India has not taken any effort to include this herb among the list of prioritized plants (Menon).

Botanical Characters

Centella asiatica (L.) is a prostrate, slender, trailing and stoloniferous herb with rooting at the nodes. It attains the height upto 15 – 20 cm. It has long prostrate stem having entire, reniform and orbicular leaves with with short peduncle, sheathing base, crenate margins and glabrous appearance on either side. Flowers are sessile, pink, covered with bracts and a group of 3 – 6 flowers remain arranged in an umbel. Flowering was mostly recorded during the month of December – March. Fruits are long, oblong and globular with thick pericarp. Seeds are laterally compressed having characteristic grayish green color.

MATERIALS & METHODS

The present study was done in the tribal dominated areas of Santhal Paragana division. This herb is commonly known as gotukola or falguni patta sag and used as vegetable by the poor mass. The tribals of this area are both socially and economically backward to be far away from the actual scene of development. If tribal communities of these areas will get support from both the Governmental and Non-governmental Organizations and will be encouraged in cultivation of this low maintenance cost medicinal plants with true and justified market value, the scene of socio-economic development will be different and it will surely change the scene of unemployment and their living standard.

RESULT & DISCUSSION

Regular survey of the sampling areas showed abundance scope of cultivation of this multifarious herbal plant on commercial and large scale. There is only need to develop proper management and secure marketing system. The leaves or whole plant are sold in local village market. Mostly fish farmers, females and girl children were found to be engaged in collection of plants.

The specific knowledge of plants as a source of medicine has been since Rig Vedic period. There are a number of secondary metabolites found in *Centella asiatica*, which play very important role in pharmacological applications. It's biologically active components are triterpenes and saponins (Loiseau & Mercier, 2000). The triterpenes found in *Centella asiatica* are asiatic acid, brahmic acid, centric acid, centoic acid, centelloic acid etc. (Zheng & Qin, 2007). Indian variety of thankuni plant contains indo – centoic acid, an isomeric form of centoic acid (Gupta & Sharma, 2007). This plant contains various kinds of volatile and fatty acids like oleic acid, linoleic acid, stearic acid etc. (Gupta & sharma 2007). An alkaloid, hydrocotylin has also been reported from ethanolic extract of *Centella* (Fanswort et al, 1976). Besides these, this herb contains a number of flavanoids, sterols, sugars and inorganic salts (Handa et al, 1998).

Ethno medicinal Properties

Centella asiatica contains high ethnopharmacological value due to its constituents and curative properties for various ailments. Some of the important pharmacological uses of *Centella* are as :

Use as Anti-oxidant

Due to presence of very high percentage of Vitamin C, *Centella* plant is well known to have high antioxidant value (Hashim et al, 2011). Indian variety of this herb is a good source of antioxidants like Ascorbic acid, Beta carotene and phenolic compounds (Gupta & Prakash, 2009). This flora has good antioxidant activity and good source of antioxidant contents. Vimala, *et al* (2003) reported that *C.asiatica* leaves were found to have very high antioxidant activity in three different pathways

including superoxide free radical scavenging activity (86.4%), inhibition of linoleic acid peroxidation (98.2%) and radical scavenging activity, DPPH (92.7%). This plant has better wound healing activity due to presence of Asiaticoside in it, which increases the antioxidant level (Shukla et al, 1999).

Antimicrobial Activity

Centella plant contains remarkable quantity of Chloroform extract, due to which it plays significant role against various microbial activities including bacteria, fungi and protozoan's. It is used frequently and successfully against various kinds of infectious diseases.

Use in Ulcer & Gastrointestinal problem

The leaves of *C.asiatica* is used in various kinds of gastrointestinal problems by the local tribal mass. People eat 5 – 10 green leaves in empty stomach daily morning and they get excellent result against gastritis and ulcer. A local herbal seller, Mangal Raut acquainted that this herb had been used in anti – ulcer activities for so many generations.

Anti diabetic effect

Centella plant contains ethanolic and methanolic extracts, due to which it shows excellent anti diabetic activity. It is used to lower down the glucose sugar in blood. It has been reported to lower down the cholesterol, triglycerides and urea levels in blood.

Neuroprotective role

It has been reported that fresh leaves of *Centella asiatica* are neuroprotective and are widely used in memory disorders and strain. It can be used as brain tonic and against anxiety and depression.

Anti-inflammatory Action

Thankuni plant is anti-inflammatory in activity due to presence of Asiatic and madecassic acid in its crude extract (Somchit et al, 2004).

Besides these pharmacological values, this elixir is cytotoxic, cardio protective, skin protective, radio protective, memory enhancing and anti allergic in action.

CONCLUSION

Thankuni plant parts has been used in traditional and alternative medicine due to very wide spectrum of pharmacological activities. It seems to be very important herb since onset of human civilization. It can be truely recommended as an elixir of human life. One of the important point is that it is economically affordable and sustainable involving minimum risks. Finally, it can be said that *C.asiatica* seems to be a potential herbal plant in many healthcare applications as well as with nutritional and cosmetic value.

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